

# Jooho Kim

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## EDUCATION

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### Seoul National University

*BS in Electrical and Computer Engineering, Mathematical Science*

- Current GPA: 3.6/4.30 ( $\approx 93.0\%$ )

Seoul, South Korea

*Mar. 2021 –*

## EXPERIENCE

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### CryptoLab, Inc. — Homomorphic Encryption Engineer

*HEaaN2 Library*

*Jun. 2026 – Present*

- Implemented encrypted MLP inference on the MNIST dataset using the HEaaN2 CKKS library (see Projects)
- Currently working on re-implementing HEaaN2 to conform with the ISO/IEC international standard
- Performed a clause-by-clause gap analysis between the ISO draft and the HEaaN2 technical specification, identifying mismatches in bootstrap pipeline ordering, EvalMod approximation, rescaling, and evaluation-key format

### Republic of Korea Army — Thermal Observation Device Operator (TOD)

*Mandatory Military Service*

*Jul. 2022 – Sep. 2023*

- Operated infrared (IR) thermal imaging systems for real-time surveillance and anomaly detection
- Designed and deployed a JavaScript-based scheduling system to automatically assign 20+ personnel across 12 daily shifts

## RESEARCH

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### Undergraduate Thesis: ISI Pre-equalization for mmWave MIMO Systems

*Jun. 2025 – Dec. 2025*

- Modeled ultra-wideband mmWave MIMO channels and inter-symbol interference using linear signal processing theory
- Designed and implemented an ISI pre-equalization scheme in MATLAB
- Ran Monte Carlo simulations to evaluate performance under realistic multipath channels
- Improved SNR performance compared to conventional equalization baselines

## PROJECTS

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### Homomorphic MNIST MLP Inference on HEaaN2 | *Homomorphic Encryption, C++* Jun. 2026 – Aug. 2026

- Implemented encrypted MLP inference on the MNIST dataset using the HEaaN2 CKKS library
- Applied PCMM (plaintext-ciphertext matrix multiplication) to efficiently evaluate matrix multiplication under encryption
- Optimized rotation and multiplication cost within the packed encoding scheme to reduce homomorphic evaluation latency

### AI Product Recognition Competition — Encouragement Award | *Deep Learning, Python*

*Jul. 2024*

- Developed a deep learning-based computer vision model for automated product recognition in unmanned retail environments
- Implemented data preprocessing, augmentation, and CNN/Transformer-based architectures
- Optimized training through hyperparameter tuning and ensemble methods
- Received Encouragement Award

## TECHNICAL SKILLS

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**Programming Languages:** C/C++, Python, JavaScript, MATLAB

**Developer Tools:** Git, LaTeX

**Languages:** Korean (Native), English (Fluent)